

# Modern Quantum Problems

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# Chap. 1 Formalism.

1.  $[AB, CD] = ABCD - CDAB$

$$= -\cancel{ACDB} - \cancel{ACBD} + \cancel{ACBD} + ABCD - CDAB - \cancel{CADB} + \cancel{CADB} + \cancel{ACDB} \quad \square.$$

2.  $X = a_0 + \sigma \cdot a$   $\sigma$  represent pauli matrix with  $\text{tr}(\sigma) = \text{I diagonals} = 0$

a.  $a_0 \quad a_k \quad (k=1,2,3) \rightarrow \text{tr}(X) \quad \text{tr}(\sigma_k X)$

$$\text{tr}(X) = \text{tr}(a_0 + \sigma \cdot (a_1 a_2 a_3)) = \sigma_2 a_2 + \sigma a_0$$

[Pauli Matrix  $\sigma$ ]:

$$X = a_0 + \sigma \cdot a = \begin{pmatrix} a_0 + a_3 & a_0 + a_1 - ia_2 \\ a_0 + a_1 + ia_2 & a_0 - a_3 \end{pmatrix}$$

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$a_0 = (X_{11} + X_{22})/2 \quad a_1 = (X_{12} + X_{21} - X_{11} - X_{22})/2$$

$$a_3 = (X_{11} - X_{22})/2 \quad a_2 = (X_{21} - X_{12})/2i$$

3.  $\sigma \cdot a' \rightarrow \sigma \cdot a$  by  $= \exp(i\sigma \cdot \hat{n} \phi/2) \sigma \cdot a \exp(-i\sigma \cdot \hat{n} \phi/2)$

$$\sigma \cdot a = \sum_i \sigma_i a_i = \begin{pmatrix} a_z & a_x - ia_y \\ a_x + ia_y & -a_z \end{pmatrix} \quad \text{Seperate } \exp(-i\sigma \cdot \hat{n} \phi/2) \text{ term by}$$

$\star e^{i\theta n \cdot \sigma} = \cos \theta I + i \sin \theta (n \cdot \sigma)$  Prof. for  $(n \cdot \sigma)^0 = 1$

$$(n \cdot \sigma)^2 = n \cdot \sigma (n \cdot \sigma) = (n \cdot n) I + n \times n i \sigma = I$$

$$(n \cdot \sigma)^1 = n \cdot \sigma$$

$$e^x = 1 + x/1! + \dots$$

$$= 1 + \frac{-i\theta n \cdot \sigma}{1} + \frac{-i\theta n \cdot \sigma^2}{2} + \dots$$

$$= 1 + \frac{-\theta^2 I}{2} + \frac{-i\theta n \cdot \sigma}{6} + \dots = \text{Re} \left[ \underbrace{1 - \frac{\theta^2}{2} + \frac{\theta^4}{4!} - \dots}_{\cos \theta} \right] I + \text{Im} \left[ i(n \cdot \sigma) \left( \underbrace{\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots}_{\sin \theta} \right) \right]$$

$$= \cos \theta I + i \sin \theta (n \cdot \sigma)$$

$$\Rightarrow \exp(i\sigma \cdot \hat{n} \psi/2) \text{ with } \theta = \psi/2 = \cos \psi/2 I + i \sin \psi/2 (n \cdot \sigma)$$

$$\text{Vice } \exp(-i\psi/2) = \cos \psi/2 I - i \sin \psi/2 (n \cdot \sigma)$$

$$\sigma \cdot a \text{ in } z+ = (\cos \psi/2 + i \sin \psi/2 \sigma_z) \begin{pmatrix} a_z & a_x - ia_y \\ a_x + ia_y & -a_z \end{pmatrix} (\cos \psi/2 - i \sin \psi/2 \sigma_z)$$

$\star$  Since  $x = a_x \sigma_x + a_y \sigma_y + a_z \sigma_z \rightarrow$  Calculate each  $\sigma$  Separately

$$= (\cos \psi/2 \cdot \sigma_x + i \sin \psi/2 \sigma_z \sigma_x) (\cos \psi/2 - i \sin \psi/2 \sigma_z)$$

$$= \cos^2 \sigma_x - i \sin \cos \sigma_z \sigma_x + i \sin \cos \sigma_z \sigma_x + \sin^2 \sigma_z^2 \sigma_x \leftarrow \text{For } \sigma_z \sigma_x = -i\sigma_y \quad \sigma_y \sigma_z = i\sigma_x$$

$$= \cos^2 \sigma_x - \sin^2 \sigma_x \quad \star \sigma_x \sigma_z \neq \sigma_z \sigma_x \quad [\sigma_x \sigma_z] \neq 0$$

$$= \cos^2 \sigma_x - \sin^2 \sigma_x + 2 \sin \cos \sigma_y = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \rightarrow \text{Rotation matrix}$$

4. a)  $XY = z|x\rangle \cdot \langle xy|x\rangle = z|x\rangle$

$$\text{Tr}(XY) = \sum_a \langle a | XY | a \rangle = \sum_{ab} \langle a | X | b \rangle \langle b | Y | a \rangle = \langle a | Y | b \rangle \langle b | X | a \rangle = \sum_a \langle YX \rangle = \text{Tr}(YX)$$

$$b). (XY)^\dagger = Y^\dagger X^\dagger \quad \langle a|(XY)^\dagger|b\rangle = \langle b|XY|a\rangle^* = \left(\sum_c \langle b|X|c\rangle^* \langle c|Y|a\rangle^*\right) = \sum_c \langle c|X^\dagger|b\rangle \langle a|Y^\dagger|c\rangle$$

$$= \sum_c \langle a|Y^\dagger|c\rangle \langle c|X^\dagger|b\rangle = Y^\dagger X^\dagger$$

c).  $\exp[i f(A)]$   $A=A^\dagger$  with  $A|a\rangle = a|a\rangle$

$$\exp(iff(\Sigma a)) = \sum \frac{(iff(a))^N}{N!} \quad \sum_a \exp[iff(a)] |a\rangle \langle a| = \sum_a \exp[iff(a)] |a\rangle \langle a|$$

$$d) \sum_a \psi_a^*(x') \psi_a(x'') = \sum_a \langle x' | a \rangle^* \langle x'' | a \rangle = \sum_a \langle x'' | a \rangle \langle a | x' \rangle = \langle x'' | x' \rangle = \delta(x'' - x')$$

|                             | *                                   | T             | + (= * + T)                         |
|-----------------------------|-------------------------------------|---------------|-------------------------------------|
| $\langle a   x   b \rangle$ | $\langle b   x^\dagger   a \rangle$ | Same          | $\langle b   x^\dagger   a \rangle$ |
| $\langle a   b \rangle$     | $\langle b   a \rangle$             | Same          | $\langle b   a \rangle$             |
| $  b \rangle$               | $\sim$                              | $\langle b  $ | $\langle b  $                       |

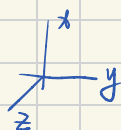
5. a.  $|\alpha\rangle |\beta\rangle$      $|\alpha\rangle\langle\beta|$      $\langle a_m|\alpha\rangle\langle\beta|a_n\rangle = \sum_{mn} \langle a_m|\alpha\rangle \cdot \langle a_n|\beta\rangle^*$

b. Spin  $\frac{1}{2}$  Intrinsic Angular Momentum: Spin  $\pm \frac{1}{2} \rightarrow \pm \frac{\hbar}{2}$

In z-basis  $|S_z = \hbar/2\rangle = |\alpha\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$$|S_8 = \hbar/2\rangle = \frac{1}{\sqrt{2}}(|+\rangle_z + |-\rangle_z) = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

★  $|S_x = \hbar/2\rangle$  in  $z$ -basis =  $1/\sqrt{2} (|z+\rangle + |z-\rangle)$



## Classical Ideas



$x, y, z$  only Canonical  
with themselves.

$$|\alpha\rangle\langle\beta| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}^\dagger = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 \end{pmatrix}$$

6.  $|4\rangle = |i\rangle + |j\rangle \rightarrow A|4\rangle = 4|4\rangle$  Also  $= a_i|i\rangle + a_j|j\rangle = a_i \text{ or } j [|i\rangle + |j\rangle]$

$\Rightarrow a_z = a_j$  degeneracy

7. a.  $\sum |a'\rangle A \prod_{a'} (A - a') |a'\rangle = 0$

$$(A - a')(A - a'') \dots |a'\rangle = 0$$

$$(A - a'')|a'\rangle = a' - 0 = A' - a' = 0$$

b.  $\prod_{a=b} (A-a)/(a-b) |a\rangle$   $f(a'') = \prod_{a'' \neq a'} \frac{(A-a'')}{(a'-a'')}$  For  $f(a'') |a'\rangle$  others = 0 Filter Operator

$$= \frac{a^{1n}}{a^{1n}} = 1$$

$$C. A = S_z \text{ Spin } \frac{1}{2} \quad (S_z - \frac{\hbar}{2}) \cdot (S_z + \frac{\hbar}{2}) = S_z^2 - \frac{\hbar^2}{4} = 0$$

8.  $|+\rangle |-\rangle$      $\langle + | - \rangle = 0$      $\hbar^2/4 (\sim)$

$$|+\rangle\langle-| + \langle+| - |+\rangle\langle-| \rightarrow \langle-| + |+\rangle\langle+| - |+\rangle\langle-| \rightarrow \langle-|$$

$$= -|+\rangle\langle-| + |+\rangle\langle+| = i\epsilon_{ijk} S_z \quad \leftarrow \quad \begin{matrix} x \rightarrow y \\ \uparrow \quad \downarrow \\ z \end{matrix} \quad \square$$

9.  $|S, \hat{n}; +\rangle$  ★ Separate term of a.b as  $a_x b_x + a_y b_y + \dots$

$$S, \hat{n} |S, \hat{n}; +\rangle = (\hbar/2) |S, \hat{n}; +\rangle \quad \cos(\beta/2) = \hbar/2$$

$$S_x \hat{n}_x + S_y \hat{n}_y + S_z \hat{n}_z = \hbar/2 (\sim) \quad \star \text{ why } S_{xyz} \text{ must belongs to Pauli Matrix}$$

$$\begin{aligned} n_x &= \sin\beta \cos\alpha \\ n_y &= \sin\beta \sin\alpha \\ n_z &= \cos\beta \end{aligned}$$

Constraints -  $[S_i, S_j] = i\hbar S_k$  (Derived From  $[x, p] = i\hbar$ )  
- Only  $\pm\hbar/2$  is possible value.

$$\text{On } S_z - \text{base} \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \hbar/2 \begin{pmatrix} 0 & 1 \\ 0 & -\hbar/2 \end{pmatrix} = \hbar/2 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$[S_z, S_x] = i\hbar S_y \quad \text{define } |+\rangle = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad |-\rangle = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Only exist these three possible matrix  $\leftarrow \Rightarrow S_x = S_+ + S_-/2 = \hbar/2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$S_y = S_+ - S_-/2 = \hbar/2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

10. ★ Observe Series of  $|1\rangle\langle 1|, |1\rangle\langle 2|, \dots$  As position in matrix.

$$\begin{pmatrix} a & a \\ a & -a \end{pmatrix} \cdot \begin{pmatrix} u \\ v \end{pmatrix} = E \begin{pmatrix} u \\ v \end{pmatrix} \quad [\hat{H}\psi = E\psi] \quad E \text{ is defined as eigenvalues of } \hat{H}$$

$$\text{with } \det(H - EI) = 0 \quad E = E = \dots \quad u, v = \dots$$

11. [Observe Paulie Matrix as base of H]

$$\begin{pmatrix} \epsilon & \delta \\ \delta & \epsilon \end{pmatrix} \text{ is Seperable to } E_{avg} \cdot I + \sigma_z \cdot E_{avg} + \sigma_x \cdot \delta$$

Unit Matrix + Values on  $\sigma_z$  + Values on  $\sigma_x$   
• Constants

$$\text{Eigenvalue of } \hat{H} = E = E_{avg} + \sqrt{\delta^2 + \delta^2} = \dots$$

$$\text{Observe eigenstate as angle operator: } \tan\theta = \text{Val. of } \sigma_z / \text{Val. of } \sigma_x = \dots$$

$$\text{with } |\hat{n}; +\rangle = \cos\beta/2 |+\rangle + \exp(i\alpha) \sin\beta/2 |-\rangle$$

12.  $xz$  plane Spin  $1/2$   $\gamma$  angle to  $\hat{z}$

★  $S_{xyz}$  Operators = Equipments on ask Self Spin on  $xyz$  direction.

$$S_{xyz} = \hbar/2 \sigma_{xyz}$$

Operating by  $|+\rangle, |+\rangle, |+\rangle$  eigenkets to find propotions.

$$S_x |\psi\rangle = \text{Useless Steps} \quad \langle \pm x | \psi \rangle = \text{P. of having } \pm\hbar/2 \text{ on } x \text{ direction.}$$

a).  $N_z = \cos\gamma/2 |+\rangle + \exp(i\alpha) \sin\gamma/2 |-\rangle$  As General Spin Operator On  $z$ -base with  $\gamma$  Angle to  $\hat{z}$   $\alpha$  Angle to  $\hat{y}$

$$\langle +x | N_z \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} \cos\gamma/2 \\ \sin\gamma/2 \end{pmatrix} = \frac{1}{\sqrt{2}} (\cos\gamma/2 + \sin\gamma/2) \Rightarrow P = \left| \frac{1}{\sqrt{2}} (\cos\gamma/2 + \sin\gamma/2) \right|^2 = \frac{1}{2} (1 + 2\sin\gamma/2 \cos\gamma/2) = \frac{1}{2} (1 + \sin\gamma)$$



b).  $\langle \Delta S_x \rangle^2 = \langle S_x^2 \rangle - \langle S_x \rangle^2$

$$\langle S_x \rangle^2 = |\langle \psi | S_x | \psi \rangle|^2 \quad \text{or} \quad = \frac{1}{2}(1+\sin\gamma)\frac{\hbar}{2} + \frac{1}{2}(1-\sin\gamma)(-\frac{\hbar}{2}) = \frac{\hbar^2}{4}\sin^2\gamma$$

### 13. Method 1 - Basic Formalism

State:  $\begin{pmatrix} a \\ b \end{pmatrix}$  Eigenkets:  $|S, \hat{n}, \pm\rangle$  Test  $\langle S, \hat{n}, \pm | \text{State} \rangle$

Z-base

i)  $\langle S, \hat{n}, + | \psi \rangle = (\cos(\gamma/2) \sin(\gamma/2)) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \cos \gamma/2$   $\psi \rightarrow \text{Eigenket} = \begin{pmatrix} \cos \gamma/2 \\ \sin \gamma/2 \end{pmatrix}$

ii).  $\langle S_z : - | \psi_{\text{new}} \rangle = (0 \ 1) \begin{pmatrix} \cos \gamma/2 \\ \sin \gamma/2 \end{pmatrix} = \sin \gamma/2$

$$P = P_1 \times P_2 = \cos^2 \gamma/2 \sin^2 \gamma/2 = \frac{1}{4} \sin^2 \gamma$$

### Method 2 - Formalism

$$P = |\langle + | - \rangle|^2 \rightarrow |\langle + | S_n | - \rangle \langle S_n | - \rangle|^2 \leftarrow \text{For } |S_n+\rangle = [\cos \gamma/2 |+\rangle + \sin \gamma/2 |-\rangle]$$

$$= \cos^2 \gamma/2 \sin^2 \gamma/2$$

$$= \frac{1}{4} \sin^2 \gamma$$

$$\hat{n} = \sin\gamma \hat{n}_x + 0 \hat{n}_y + \cos\gamma \hat{n}_z$$

$$S \cdot \hat{n} = \begin{pmatrix} \cos\gamma & \sin\gamma \\ \sin\gamma & -\cos\gamma \end{pmatrix} \cdot \frac{\hbar}{2}$$

$$|S \cdot \hat{n}, \pm\rangle = \frac{\hbar}{2} \begin{pmatrix} \cos\gamma & \sin\gamma \\ \sin\gamma & -\cos\gamma \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = \pm \frac{\hbar}{2} \begin{pmatrix} a \\ b \end{pmatrix}$$

Only ratio needed

$$\Rightarrow \cos\gamma \cdot a + \sin\gamma \cdot b = a$$

$$a = \left( \frac{\sin\gamma}{1-\cos\gamma} \right) b = \frac{2\sin\gamma/2 \cos\gamma/2}{1-1+2\sin^2\gamma/2} b = \cos\gamma/2 / \sin\gamma/2 \cdot b$$

$$|S \cdot \hat{n} : +\rangle = \begin{pmatrix} \cos \gamma/2 \\ \sin \gamma/2 \end{pmatrix}$$

Similar

$$|S \cdot \hat{n} : -\rangle = \begin{pmatrix} \sin \gamma/2 \\ \cos \gamma/2 \end{pmatrix}$$

— Vector

— Operator

★ Logic Chain of Vector

Eigenkets ← Operator

← Eigenkets of  $S \cdot \hat{n}$

$$\text{Set } a = \cos \gamma/2 \quad b = \sin \gamma/2$$

— Eigenkets



14. a.  $\det(A - \lambda I) = 0$

$$(-\lambda^3) - 2(-\lambda)(1/\sqrt{2})^2 = 0 \quad \lambda = \pm 1, 0$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = \sqrt{2} \begin{pmatrix} a \\ b \end{pmatrix} \quad \begin{matrix} 2 \\ b = \sqrt{2}a \end{matrix} \quad a+c = \sqrt{2}b \quad a=c$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{2} \\ 2 \\ \sqrt{2} \end{pmatrix}$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{2} \\ -2 \\ \sqrt{2} \end{pmatrix}$$

$$15. \hat{A}\hat{B} \quad \{|a', b'\rangle\} \quad [A, B] = 0$$

$$\begin{aligned} [A, B]|ab\rangle &= AB|ab\rangle - BA|ab\rangle = A b' |a', b'\rangle - B a' |a', b'\rangle \\ &= b' a' |a', b'\rangle - a' b' |a', b'\rangle = 0 \end{aligned}$$

$$[A, B]\psi = \sum C [A, B]|ab\rangle = 0.$$

$$16. \{A, B\} = AB + BA = 0 \quad |a, b\rangle$$

$$2ab|ab\rangle = 0 \quad a \text{ or } b \text{ or } a, b = 0.$$

$$17. A_1, A_2 \quad [A_1, A_2] \neq 0 \quad [A_1, H] = [A_2, H] = 0$$

$$H|E\rangle = E|E\rangle \rightarrow A_1 H|E\rangle = H A_1 |E\rangle$$

$$\begin{array}{ccc} \underline{E_2 A_1}| \neq \underline{E_1 A_2}| & \leftarrow & \cancel{A_1}| = \cancel{A_2}| \\ \downarrow & & \downarrow \\ E_2 = E_1 & & \end{array}$$